CoHERence

Understanding of the problem domain

- Urban and policy infrastructure carry male-default design bias - planning systems built without gender-disaggregated data.
- Public facility provisioning shows spatial gender-blind spots - no correlation between women's workplace geography and essential service access.
- Sector documents and legal drafts harbor implicit linguistic bias, largely undetected due to lack of systematic bias auditing.
- Absence of consolidated gender-disaggregated workforce data limits evidence-based policy targeting.
- ML systems trained on this same biased data risk algorithmic bias amplification in welfare, hiring, and credit-scoring pipelines.

Proposed Solution

- A geospatial platform that maps where women work and cross-references it against essential public infrastructure - safety, childcare, healthcare, transit.
- Flags facility deserts: zones with high female workforce presence but low access to essential services.
- A workforce data layer that consolidates gender-disaggregated data - tracking working, educated, and non-working women across regions and sectors.
- Together, the two layers turn scattered, invisible gaps into a single, prioritized, data-backed view - showing exactly where and what to fix first.

Relevance to the Track

- A literal build of the track's premise - not referenced in spirit, but operationalized as the redesign instrument itself, applied to the systems shaping her daily life.
- Facility Gap Mapping answers the spatial claim directly - geospatial, block-by-block visibility into where infrastructure wasn't built around her routine.

- Workforce Data Dashboard redesigns the data layer - surfacing gender-disaggregated metrics on working, educated, and unemployed women who existed but were never tallied.
- Together, both modules mean the solution doesn't patch one symptom - it redesigns the structural layers where the gap was hiding, matching the track's exact mandate.

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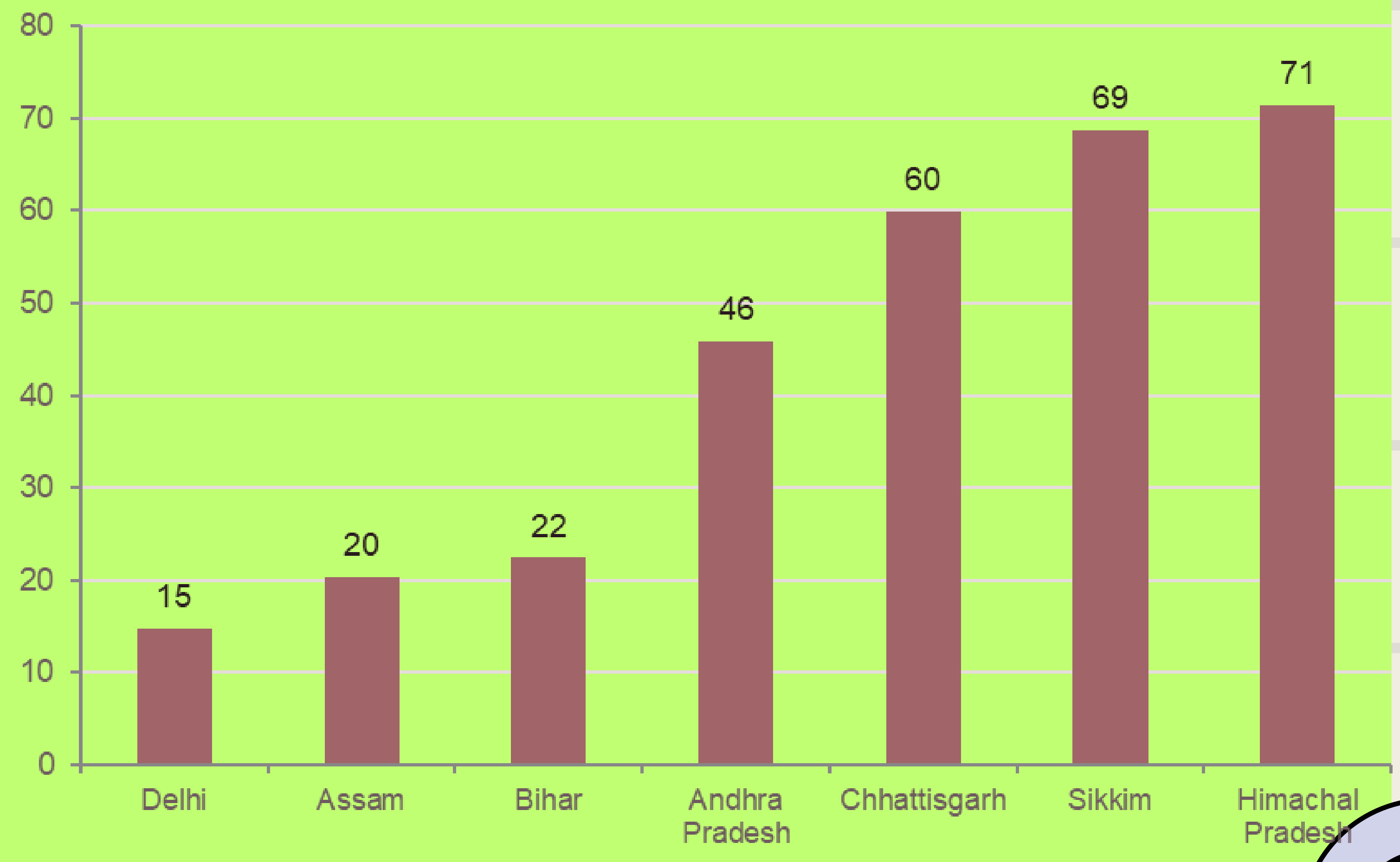
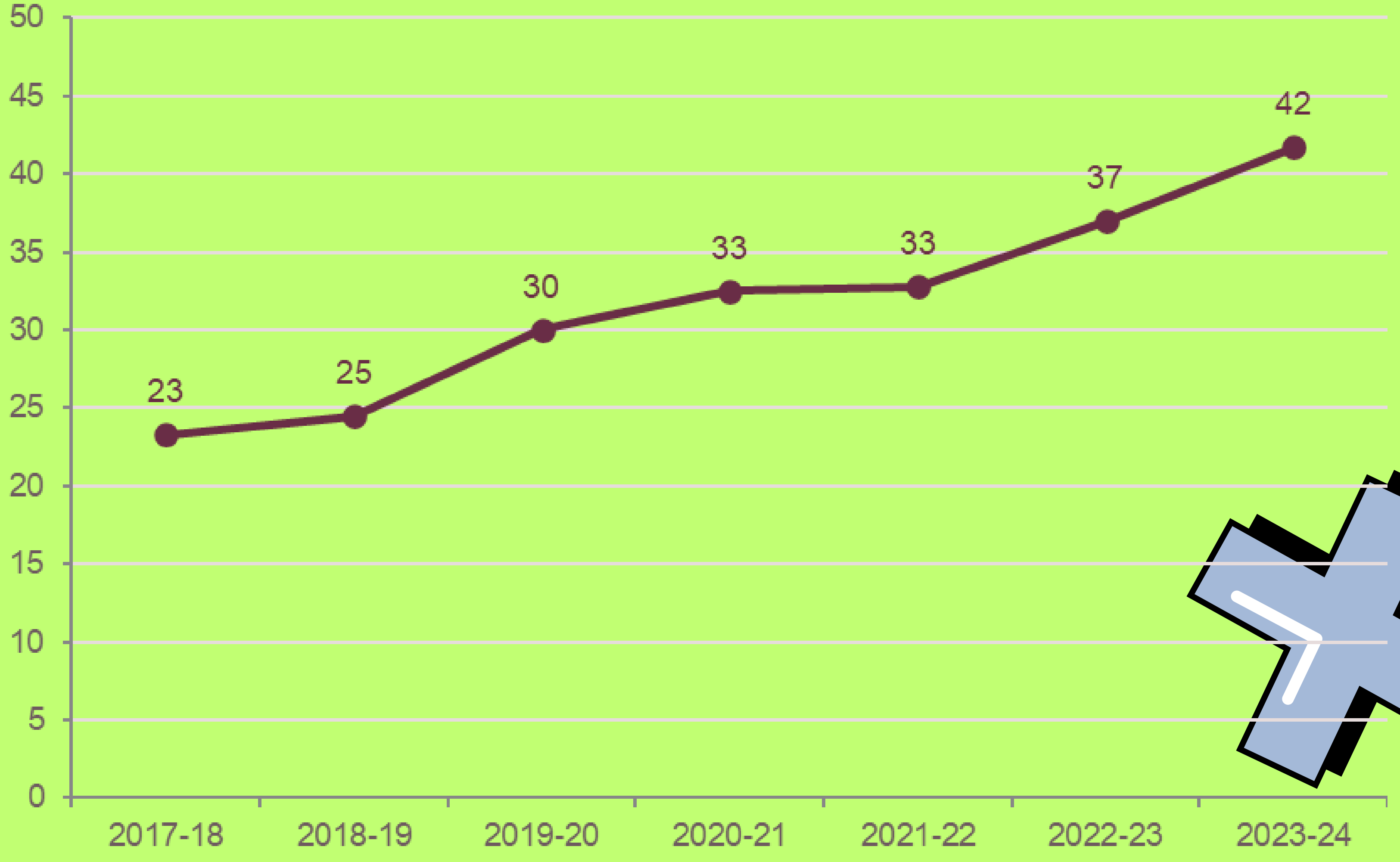
India's female workforce participation varies from state to state





India's female labour force participation rate in percentage

Innovation and originality

- Cross-domain integration: unifies spatial equity mapping (GIS) with gender-disaggregated workforce analytics into a single operational pipeline - most existing tools address these in isolation.
- Closed-loop design: facility gaps identified spatially connect directly to workforce data, turning fragmented observations into one prioritized, actionable view.
- Reframes a qualitative problem as measurable: converts "the world isn't built for her" from an anecdotal, felt experience into a quantifiable, trackable gap score.
- Built for adoption, not just demonstration: designed to plug into existing public data systems rather than function as a standalone prototype - so impact scales with reuse, not just relevance.

Feasibility

- Roads, transit, footpaths: OpenStreetMap (Geofabrik/QuickOSM extracts) - strong coverage for Indian cities.
- Facility layers (toilets, hospitals, streetlights): OSM tags + VIIRS Nighttime Lights for lighting gaps; regional open-data portals for added depth.
- Workforce data: PLFS + Census microdata - public datasets providing employment, education, and workplace-location fields at district level.
- Pedestrian density: no direct open layer - modeled via OSM footway density + WorldPop + night-lights as a proxy.
- Gap acknowledged: safety/surveillance-related data is largely non-public; approached via proxy indicators (facility density, lighting coverage) instead.

Potential Impact

- Individual-level impact: fewer invisible risk calculations, detours, and safety trade-offs baked into a woman's daily commute.
- Retention impact: closes the gap between a woman staying in the workforce or dropping out due to unsafe or exhausting daily conditions - directly tied to female labour force participation.
- Scalability: fully GIS-based and interoperable with existing public data systems - any district or state with facility data can run the same gap analysis, so impact compounds with adoption.
- Evidence-based governance: replaces guesswork with real, gender-disaggregated numbers on working, educated, and non-working women.
- Structural shift: moves from reactive, anecdotal response to default, structural planning - the same standard already applied to everyone else - making the track's redesign mandate measurable and actionable.